PART OF nocn GROUP

Functional Skills
Mathematics Assessment
Level 2

There are 2 sections in this assessment, Part A and Part B.



Learner name

_____

Run ID

_____

Learner signature

_____

Centre

_____

Assessment date

_____

Part A - Non-calculator

You have 30 minutes to complete this section.

You must not use a calculator.

asses.

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

60 / 60 Marks (100% Pass · Full Marks)

- Assessment: NOCN Functional Skills Mathematics Level 2 Sample 1
- Complete stepped calculations, diagrams & mark scheme allocations [MI/AI]
- Exam Board: NOCN (National Open College Network · Ofqual Regulated)
- Pass requirement: 36 / 60 marks (Achieved: 60 / 60 marks)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

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NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 1

2456 + (–563) =

Adding a negative number is equivalent to subtraction:

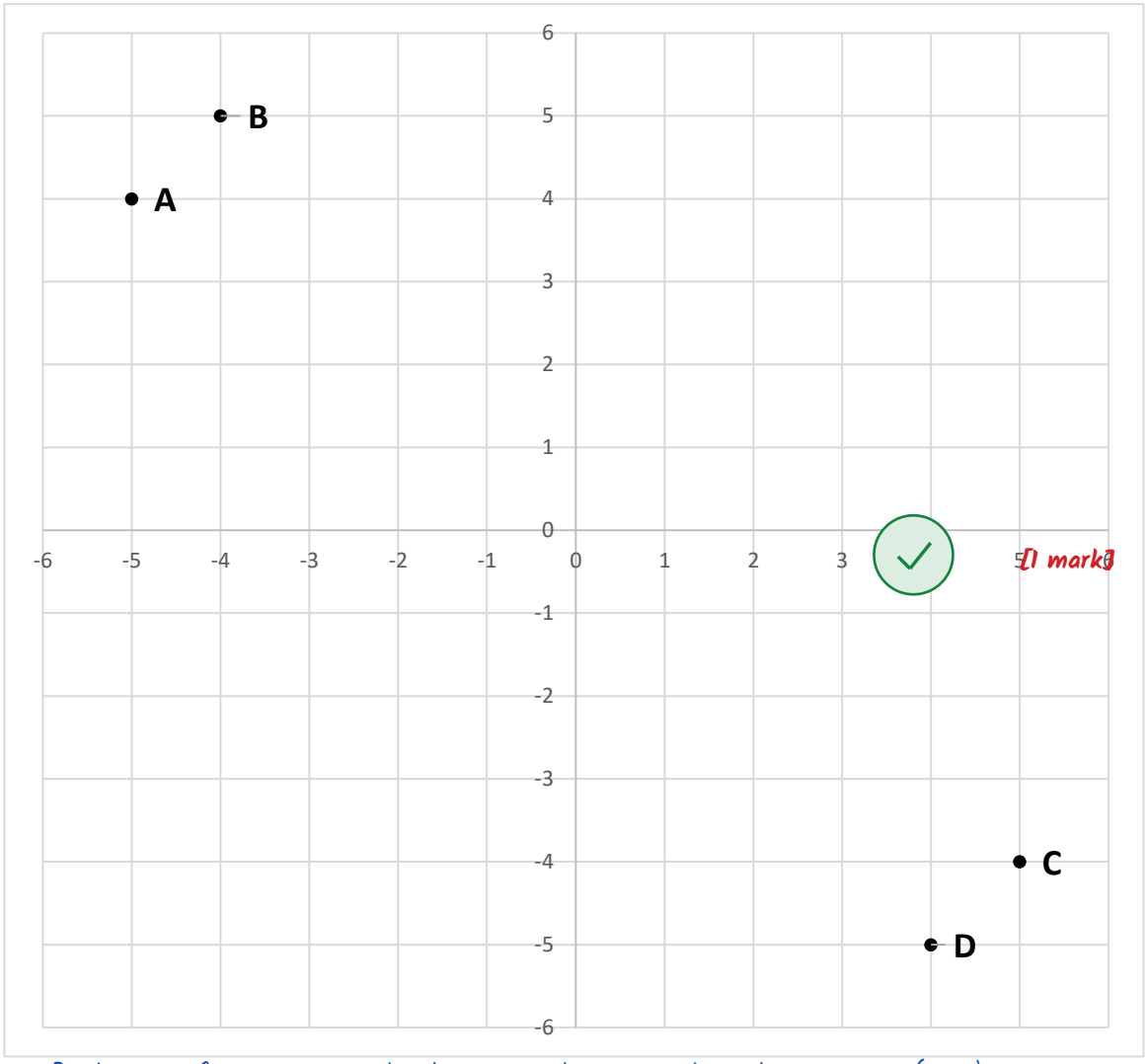
Show your working out and write your answer in the box below. **(1 mark)**

2456 + (-563) = 2456 - 563 = 1893 *[1 mark]*

Answer 1893

NOCN Functional Skills Mathematics Level 2 – Reform Sample Paper 1 Assessment Paper

Question 2



Reading point C: move 5 units right along x-axis, then 4 units down along y-axis --> (5, -4).

What are the coordinates of C? Tick ✓ the correct box. **(1 mark)**

1) (-4, 5) ☐

2) (5, -4) ☐

3) (4, -5) ☐

4) (-5, 4) ☐

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

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Question 3

Find the mode of this data.

36400g 54.2kg 36.4kg 56kg 22.1kg 4300g 51.5kg 75kg

Show your working out and write your answer in the box below. **(2 marks)**

Step 1: Convert all weights to kilograms (÷ 1000):

36400 g = 36.4 kg and 4300 g = 4.3 kg *[1 mark]*

Step 2: List all weights in kilograms:

36.4, 54.2, 36.4, 56.0, 22.1, 4.3, 51.5, 75.0 kg

Step 3: The mode is the most frequently occurring value:

36.4 kg appears twice; all other values appear only once. *[1 mark]*

Answer **36.4 kg (or 36,400 g)** ✓

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

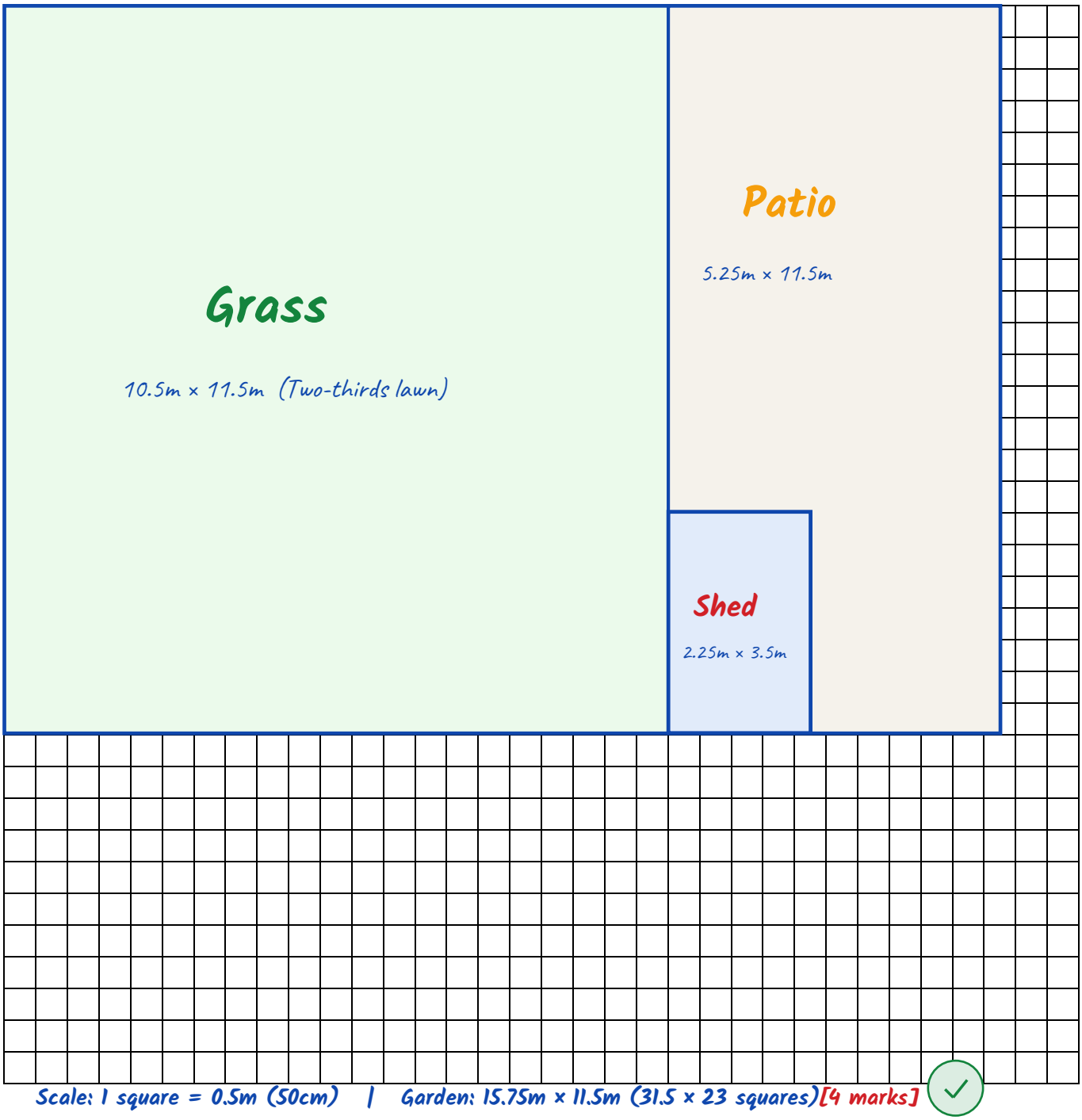
Question 4

Tim has decided to landscape his garden. His garden is a rectangle 11.5m by 15.75m.

He wants to have two thirds grass and the rest as a patio. He also wants a shed 3.5m x 2.25m to be placed in any corner of the patio.

Draw a scale drawing on the graph paper below, label the drawing clearly.

(4 marks)



Scale: 1 square = 0.5m (50cm) | Garden: 15.75m × 11.5m (31.5 × 23 squares) [4 marks]

Question 5

Step 1: Find common denominator (lowest common multiple of 6 and 4 is 12):

$\frac{5}{6} + \frac{13}{4} =$ 5/6 = (5 x 2) / (6 x 2) = 10/12 [1 mark]

13/4 = (13 x 3) / (4 x 3) = 39/12

Show your working out and write your answer in the box below. **(2 marks)**

Step 2: Add numerators: 10/12 + 39/12 = 49/12 = 4 1/12 [1 mark]

Answer 49 / 12 (or 4 1/12) ✓

Question 6

Allen is buying a house, it costs 1 million pounds.
He has a deposit of £378,500. The remaining amount he will borrow from the bank.

How much will Allen borrow from the bank?

Step 1: Subtract deposit paid from total purchase price:

Show your working out and write your answer in the box below. **(2 marks)**

£1,000,000 - £378,500 = £621,500 [2 marks]

Verification: £621,500 + £378,500 = £1,000,000 (correct).

Answer £621,500 ✓

Question 7

Calculate the area of a triangle where height = 15cm and base = 7cm.

Step 1: Area of right-angled triangle = 1/2 × base × height

Show your working out and write your answer in the box below. **(2 marks)**

Area = 1/2 × 7 cm × 15 cm = 105 ÷ 2 = 52.5 cm² [2 marks]

Always state square units (cm²) for full marks.

Answer 52.5 cm² ✓

Question 8

What is the probability of picking a blue ball from a box with 16 green, 4 purple and 5 blue? Write your answer as a percentage to 2dp.

Step 1: Calculate total number of marbles

Show your working out and write your answer in the box below. **(2 marks)**

Total = 16 red + 4 blue + 5 green = 25 marbles [1 mark]

Step 2: Fraction of green marbles = 5 / 25

Step 3: Convert to percentage: (5 ÷ 25) × 100 = 20% [1 mark]

Answer 20% ✓

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

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End of Part A
Non-calculator

The invigilator will collect this section when the assessment time has elapsed.

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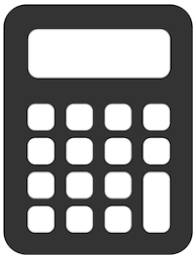
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NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

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Functional Skills
Mathematics Assessment
Level 2



Learner name

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Part B - Calculator

You have 1 hour 45 minutes to complete this section.

You <u>may</u> use a calculator.

Instructions to learners

Part B: Calculator

You will need a pen, pencil and a ruler. You may need a protractor and compasses.

Use only blue or black ink. Pencil **<u>must only</u>** be used for drawing.

You can write or draw to show your answers.

You have **1 hour 45 minutes** to complete **Part B**.

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 9

Trevor wants to buy a new lorry for work.
Trevor has to transport milk from the farms.



He will need a lorry that is suitable to carry liquid.

The liquid will be carried in a cylinder.

He has seen two lorries for sale:

Lorry 1 has a capacity of 536ft^3

Lorry 2 has the following dimensions: length 19.35ft and diameter 7.6ft

π= 3.14

Trevor wants to buy the larger lorry. Which should he buy? Show clearly how you decided.

Step 1: Calculate radius of tank = Diameter ÷ 2 = 7.6 ft ÷ 2 = 3.8 ft [1 mark]

Step 2: Formula for cylinder volume = π × r^2 × h (using π = 3.14):

Show your working out and write your answer in the box below. **(4 marks)**

Volume = 3.14 × $(3.8)^2$ × 19.35 = 3.14 × 14.44 × 19.35 [1 mark]

Volume = 877.36 ft^3 [1 mark]

Step 3: Compare capacities: Lorry 1 = 877.36 ft^3 vs Lorry 2 = 1040 ft^3

1040 ft^3 > 877.36 ft^3. Lorry 2 has the greater capacity. [1 mark]

Answer Lorry 2 (capacity 1040 ft^3 > 877.36 ✓) Lorry _____

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 10

Dahlia works 21 hours a week.

Her employer says she will be able to increase her hours by 3 in a month's time.

What will be the percentage increase in Dahlia's hours? Round to the nearest whole number.

Step 1: Increase in hours = 24 - 21 = 3 hours

Show your working out and write your answer in the box below. **(3 marks)**

Step 2: Percentage increase = (Increase ÷ Original) × 100

Percentage increase = (3 ÷ 21) × 100 = 14.2857...% *[1 mark]*

Step 3: Round to nearest whole number = 14% *[2 marks]*

Answer **14%** ✓

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 11

Terri has read that the span of a person's arm is correlated with their height.

He decides to do an investigation to prove this theory.

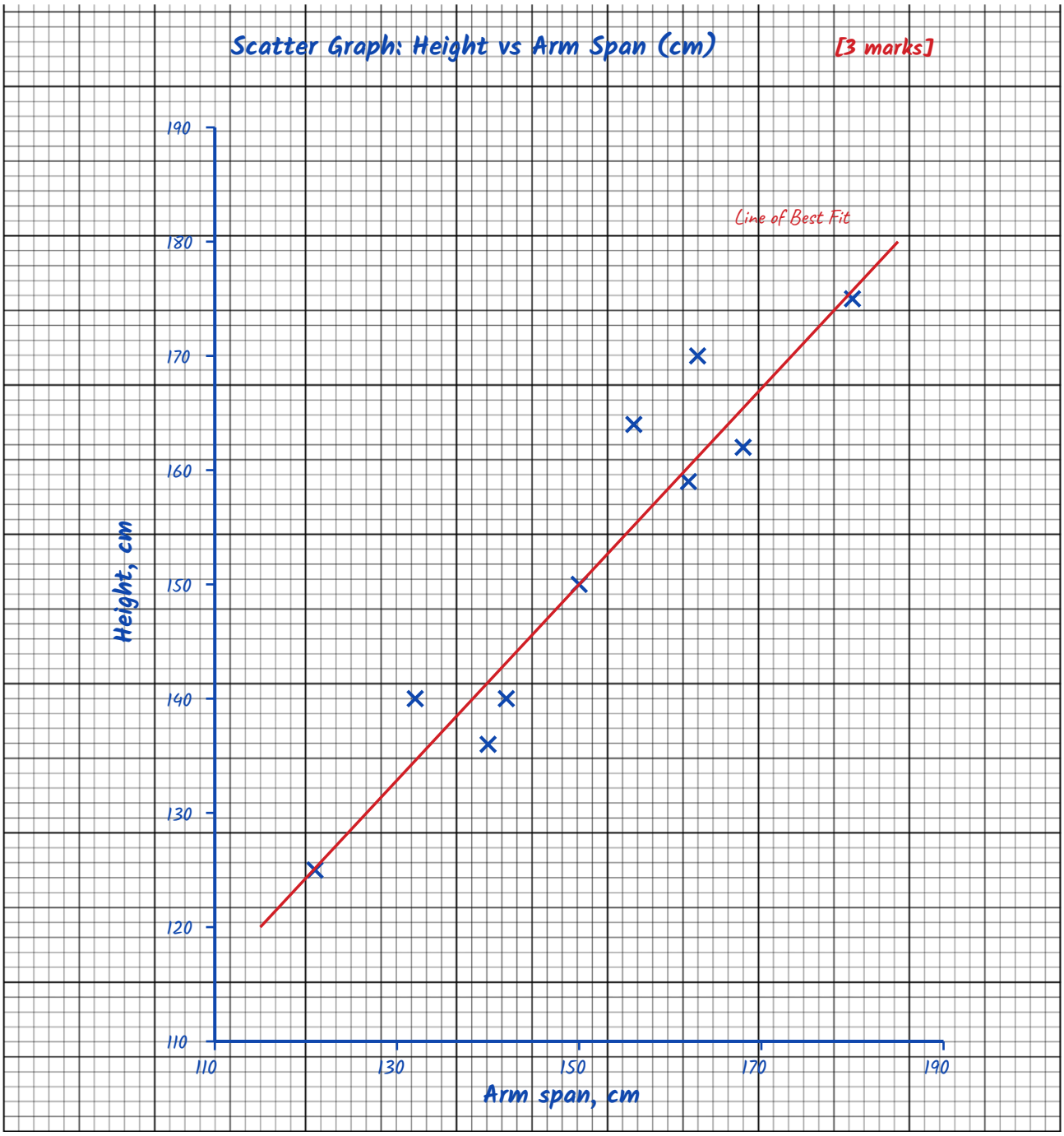
He measures the arm span and height of 10 people and records the information in a table.

Person	Arm span cm	Height cm
1	121	125
2	140	136
3	132	140
4	150	150
5	156	164
6	168	162
7	163	170
8	180	175
9	142	140
10	162	159

Draw a diagram on the next page to show the data supplied.

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Use this graph paper to draw your diagram. **(3 marks)**



Describe the correlation the data displays.
Write your answer in the box below. **(1 mark)**

As arm span increases, height also increases (strong positive correlation).

The plotted points follow a clear upward linear trend. [1 mark]

Answer Positive correlation ✓

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 12

Harry cycles on an indoor track.
The record for distance travelled on an indoor track in an hour is 53.731km.

Harry cycled 41.829km in 47.75 minutes.
Harry thinks if he continued at the same speed he could beat the record.

Step 1: Convert time into decimal minutes:

Could Harry beat the record? Show why you think this.

47 mins 45 secs = 47 + (45/60) = 47.75 minutes

Show your working out and write your answer in the box below. **(3 marks)**

Step 2: Calculate Harry's average speed in km/h:

Speed = 41.829 km ÷ 47.75 mins x 60 = 52.56 km/h [2 marks]

Step 3: Distance covered in 1 hour = 52.56 km (vs record 53.731 km):

Shortfall = 53.731 - 52.56 = 1.171 km. No, he would not beat the record. [1 mark]

Answer No, covers 52.56 km (short by 1.17 ✓)

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 13

Dave is organising a charity event for his company. The company will increase the total raised by staff by 23.2%.

Dave has collected £321.45, which is $\frac{3}{5}$ of the money promised by staff.

How much will be raised in total?

Show your working out and write your answer in the box below. **(4 marks)**

Step 1: Calculate average raised per week: £321.45 ÷ 3 = £107.15 [1 mark]

Step 2: Total raised over 5 weeks: £107.15 × 5 = £535.75 [1 mark]

Step 3: Company matching contribution (23.2%):

Match = £535.75 × 0.232 = £124.294 [1 mark]

Step 4: Total sum raised = £535.75 + £124.29 = £660.04 [1 mark]

Answer £660.04

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 14

An Ice Hockey League records the scores for the top two teams after 10 matches.

Team A Goals Scored per match	Team B Goals Scored per match
41	29
42	51
55	32
61	43
27	32
19	57
30	59
44	41
49	33
55	40
Range: 42	Range:
Mean: 42.3	Mean:

The coach of Team A says Team A scored more goals on average and were the most consistent goal scorers.

Is the coach correct? Show how you know this.

Show your working out and write your answer in the box below. **(4 marks)**

Step 1: Team B Range = Highest − Lowest = 59 − 29 = 30 [1 mark]

Step 2: Team B Mean = (29+51+32+43+32+57+59+41+33+40) ÷ 10 = 417 ÷ 10 = 41.7 [1 mark]

Consistency: Team B was more consistent (lower range: 30 < 35). [1 mark]

Goals scored: Team A scored more on average (higher mean: 44 > 41.7). [1 mark]

Answer Team B more consistent (range 30 < 35); Team A higher mean (44 > 41.7)

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 15

Sonia works in a call centre. She wants to calculate the average length of a call.

She records the following information:

Length of call (mins)	Frequency
0 - 2	40
3 - 5	32
6 - 8	30
9 - 11	25
12 - 14	20

Estimate the mean length of a call.

Show your working out and write your answer in the box below. **(5 marks)**

Step 1: Calculate class midpoints: 1, 4, 7, 10, 13 [1 mark]

Step 2: Multiply midpoints by frequencies (fx):

1 × 40 = 40 | 4 × 32 = 128 | 7 × 30 = 210 [1 mark]

10 × 25 = 250 | 13 × 20 = 260

Step 3: Total fx = 40 + 128 + 210 + 250 + 260 = 888 [1 mark]

Step 4: Total frequency = 40 + 32 + 30 + 25 + 20 = 147 [1 mark]

Step 5: Estimated Mean = 888 ÷ 147 = 6.0408... --> 6.04 minutes [2 mark]

Answer 6.04 minutes ✓

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 16

Tyler is driving from London to Plymouth.

Tyler travels at an average speed of 40mph covering 32.1 miles to the gallon.

Petrol at his local petrol station is currently 128.5p per litre.

Mileage chart (miles)

3.785 litre to 1 gallon

	Liverpool	London	Manchester	Newcastle	Norwich	Nottingham	Oxford	Penzance	Plymouth	Sheffield	Southend
Liverpool		216	34	167	222	107	168	353	299	74	255
London	216		200	278	114	130	55	282	214	161	43
Manchester	34	200		136	177	68	157	324	281	40	225
Newcastle	167	278	136		252	159	253	481	412	133	299
Norwich	222	114	177	252		118	159	393	327	144	99
Nottingham	107	130	68	159	118		104	323	255	37	160
Oxford	168	55	157	253	159	104		261	193	137	105
Penzance	353	282	324	481	393	323	261		75	361	307
Plymouth	299	214	281	412	327	255	193	75		293	269
Sheffield	74	161	40	133	144	37	137	361	293		197
Southend	255	43	225	299	99	160	105	307	269	197	

How much will the petrol cost and how long will it take to get to Plymouth in hours and minutes?

Show your working out and write your answer in the box below. **(6 marks)**

Step 1: Distance = 214 miles. Gallons needed = 214 ÷ 32.1 = 6.6667 gal [1 mark]

Step 2: Litres needed = 6.6667 × 3.785 = 25.2334 litres [1 mark]

Step 3: Petrol cost = 25.2334 × 128.5p = 3242.49p = £32.42 [1 mark]

Step 4: Journey time = Distance ÷ Speed = 214 ÷ 40 = 5.35 hours [1 mark]

Step 5: Convert 0.35 hrs to mins: 0.35 × 60 = 21 mins --> 5 hrs 21 mins [1 mark]

Answer *Cost: £32.42 | Time: 5 hours 21 minutes* ✓

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 17

Alex wants to invest £14000 in a savings account for 2 years.

He finds information about two savings accounts. Both accounts pay interest at the end of each year.

Saver 1	**Saver 2**
3% each year	5.3% for the first year 0.9% for each extra year

Saver 1 (3% compound interest each year):

Alex thinks he will earn more interest at the end of 2 years with Saver 1.

Year 1: £14,000 × 1.03 = £14,420 | Year 2: £14,420 × 1.03 = £14,852.60 [2 marks]

Is Alex correct? Explain your answer.

Saver 2 (5.3% year 1, then 0.9% extra year):

Show your working out and write your answer in the box below. **(5 marks)**

Year 1: £14,000 × 1.053 = £14,742 | Year 2: £14,742 × 1.009 = £14,874.68 [2 marks]

Decision: Alex is incorrect. Saver 2 pays £22.08 more interest

(£14,874.68 vs £14,852.60; interest £874.68 vs £852.60). [1 mark]

Answer Alex is incorrect: Saver 2 pays £22.08 more (£14,874.68 vs £14,852.60)

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

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Question 18

Zeb received a student loan of £12780 last year. Her student accommodation cost £142 per week for 30 weeks.

What fraction of her student loan was used to pay for student accommodation? Write the fraction in its simplest form.

Show your working out and write your answer in the box below. **(2 marks)**

Step 1: Calculate total accommodation cost for 30 weeks:

Cost = £142 × 30 = £4,260

Step 2: Express as a fraction of £12,780 student loan: 4260 / 12780 [1 mark]

Step 3: Simplify fraction: 4260 ÷ 4260 = 1 and 12780 ÷ 4260 = 3 [1 mark] > 1 / 3

Answer 1 / 3

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 19

Markus hires a car for 8 days for a work trip. He travels 378 miles.

The car hire company use the following formula to calculate the total hire cost:

Hire Cost = 27d + 0.2m
d = Number of Days
m = Miles Travelled

Markus can claim £300 in expenses for car hire.

Will this cover the cost of the car hire? Explain your answer.

Show your working out and write your answer in the box below. **(3 marks)**

Step 1: Substitute d = 8 days and m = 378 miles into hire formula:

Hire Cost = 27d + 0.2m = (27 × 8) + (0.2 × 378) *[1 mark]*

Hire Cost = £216 + £75.60 = £291.60 *[1 mark]*

Step 2: Compare with £300 expenses budget: £291.60 < £300 (under by £8.40).

Yes, his budget will cover the total hire cost. *[1 mark]*

Answer **Yes, £291.60 is within budget (under by £8.40)** ✓

Question 20

92(45 ÷ 5 x 3) + 32 =

Write your answer in the box below. **(1 mark)**

Step 1: Brackets: 45 ÷ 5 × 3 = 9 × 3 = 27

Step 2: Multiply: 92 × 27 = 2484

Step 3: Add: 2484 + 32 = 2516 *[1 mark]*

Answer **2516** ✓

NOCN Functional Skills Mathematics
Level 2 – Reform Sample Paper 1
Assessment Paper

Question 21

Rona is a member of a fishing club. She records the weights of the fish she catches in one day.

Fish	Weight(g)
1	4356
2	4851
3	4985
4	4913
5	4803

The heaviest fish caught by a club member was 10.8lbs.

1lb = 0.454kg

What is the percentage difference between the median weight of fish Rona caught and the heaviest fish caught by a fishing club member?

Step 1: Find median weight of 5 caught fish:

Order weights: 4356, 4803, 4851, 4913, 4985 g --> Median = 4851 g [1 mark]

Show your working out and write your answer in the box below. **(4 marks)**

Step 2: Convert record fish weight into grams (1 lb = 0.454 kg):

10.8 lbs × 0.454 kg/lb = 4.9032 kg = 4903.2 g [1 mark]

Step 3: Difference in weight = 4903.2 g − 4851 g = 52.2 g [1 mark]

Step 4: Percentage difference = (52.2 ÷ 4903.2) × 100 = 1.0646...% --> 1.06% [1 mark]

(Alternative calculation: 52.2 ÷ 4851 × 100 = 1.08%)

Answer Median: 4851 g | Percentage diff: 1.06% ✓

End of Part B
Calculator

The invigilator will collect this section when the assessment time has elapsed.

End of Assessment.

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Functional Skills
Mathematics Assessment
Level 2
Marking Sheet

Question	Available Marks	1st Marker	2nd Marker
Q1	1	1	
Q2	1	1	
Q3	2	2	
Q4	4	4	
Q5	2	2	
Q6	2	2	
Q7	2	2	
Q8	2	2	
Q9	4	4	
Q10	3	3	
Q11	4	4	
Q12	3	3	
Q13	4	4	
Q14	4	4	
Q15	5	5	
Q16	6	6	
Q17	5	5	
Q18	2	2	
Q19	3	3	
Q20	1	1	
Q21	4	4	
Total	**64**	64	

Marker's name and date

Tutor Abdul (Senior Assessor)

Moderator's name and date (if sampled)

Official Exemplar Pass (64/64)

Comments from Marker

Exemplary Level 2 model solution paper. All complex calculations, scatter plot,

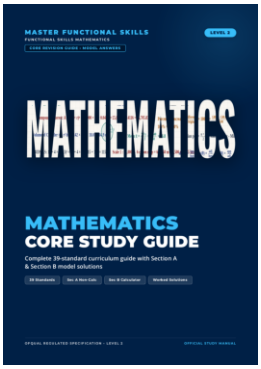
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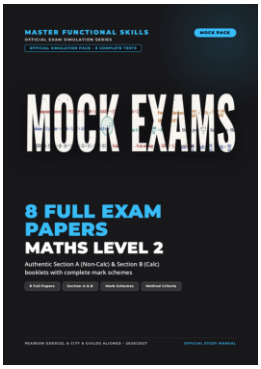


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